

Somaiya Vidyavihar University K J
Somaiya School of Engineering

8+8+4+4 = 24/25

Batch: A2 Roll No.: 16010324044

Experiment / assignment / tutorial No. 02

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff in-charge with date

Title: Frequency Shift Keying

Objectives:

1. To implement Frequency Shift Keying (FSK) digital modulation technique (transmitter)
2. To write a program for FSK in MATLAB

OUTCOME: analyze different band pass digital transmission and reception of signal

Draw Circuit Diagram:

Procedure:

- Assemble the circuit as shown in circuit diagram
- Apply digital modulating signal and analog carrier signals with given frequency and amplitude.
- Observe the FSK output on DSO and tabulate the readings.
- Draw the waveforms of modulating, carrier and FSK signal.
- Observe the waveforms of FSK on spectrum analyser.
- Write a program in MATLAB to observe the FSK waveforms.

Observation Table:

Modulating Signal		Carrier Signal			BFSK Signal	
Amplitude	Frequency	Amplitude	Frequency 1	Frequency 2	Frequency for Logic 0 input	Frequency for Logic 1 input
5V	1kHz	5V	4.5/5 kHz	13 kHz	13 kHz 4.5 kHz	4.5/5kHz 13 kHz

Matlab Code:

```
clc;
clear;
close all;
%% Input Binary Data
data = [1 0 1 1 0 0 1 0];
%% Parameters
A = 1;
f1 = 10;           % Frequency for Logic 1
f0 = 5;           % Frequency for Logic 0
Tb = 1;           % Bit duration
fs = 1000;        % Sampling frequency
t = 0:1/fs:Tb-1/fs;
%% Generate NRZ Signal
```

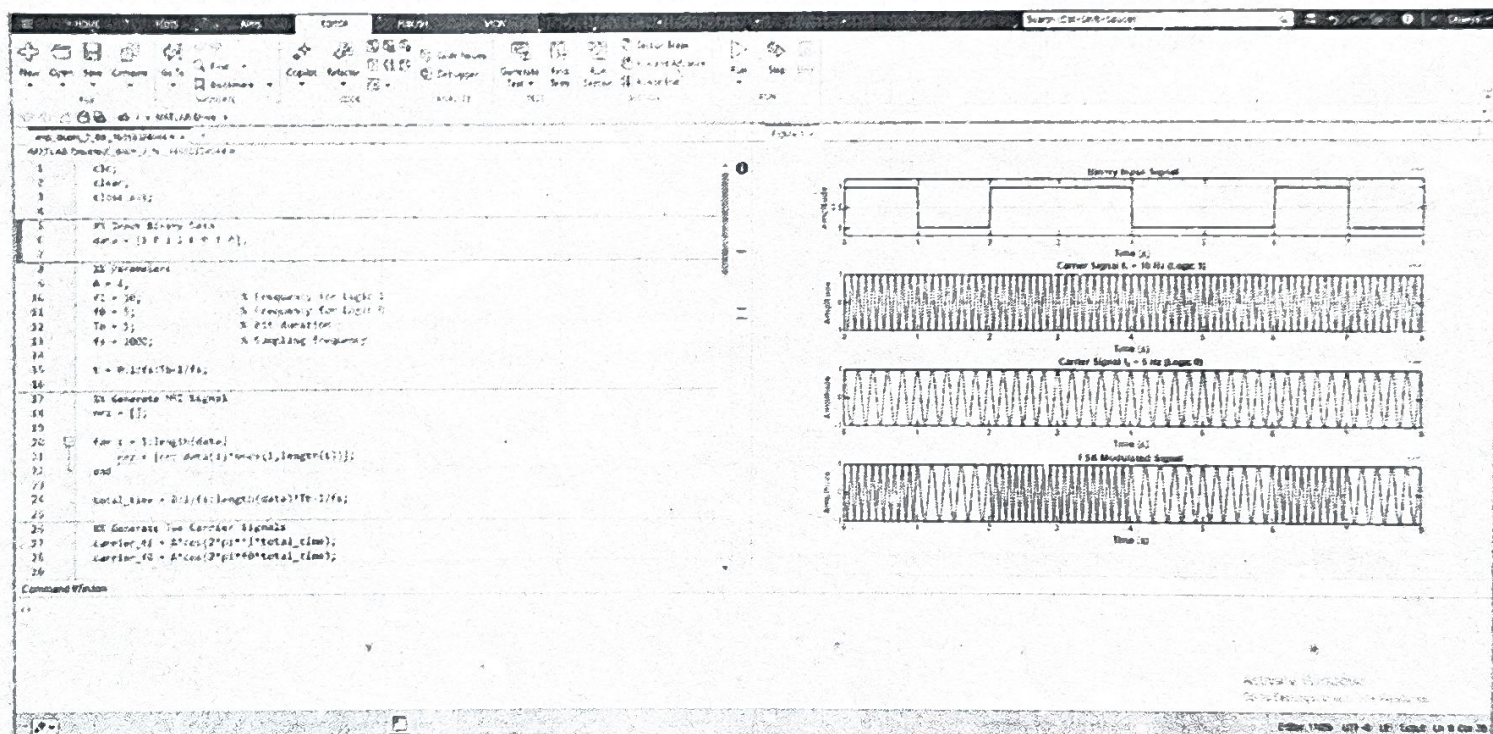
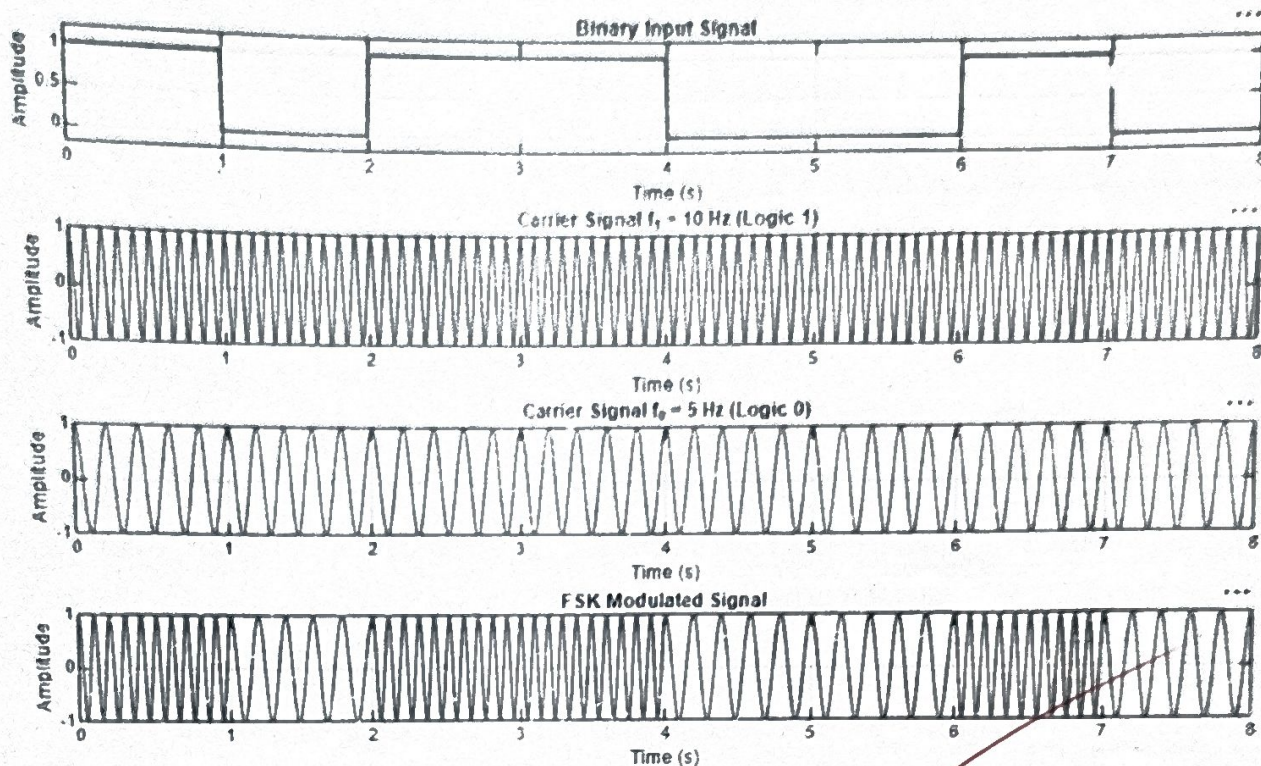
```

nrz = [];
for i = 1:length(data)
    nrz = [nrz data(i)*ones(1,length(t))];
end
total_time = 0:1/fs:length(data)*Tb-1/fs;
%% Generate Two Carrier Signals
carrier_f1 = A*cos(2*pi*f1*total_time);
carrier_f0 = A*cos(2*pi*f0*total_time);
%% Generate FSK Signal
fsk = [];
for i = 1:length(data)
    if data(i) == 1
        % Logic 1 → f1 = 10 Hz
        fsk = [fsk A*cos(2*pi*f1*t)];
    else
        % Logic 0 → f0 = 5 Hz
        fsk = [fsk A*cos(2*pi*f0*t)];
    end
end
end
%% Plot Signals
figure;
% 1. Binary Input
subplot(4,1,1)
plot(total_time,nrz,'LineWidth',2)
axis([0 length(data) -0.2 1.2])
grid on
title('Binary Input Signal')
xlabel('Time (s)')
ylabel('Amplitude')
% 2. Carrier f1
subplot(4,1,2)
plot(total_time,carrier_f1,'LineWidth',1.2)
grid on
title('Carrier Signal f_1 = 10 Hz (Logic 1)')
xlabel('Time (s)')
ylabel('Amplitude')
% 3. Carrier f0
subplot(4,1,3)
plot(total_time,carrier_f0,'LineWidth',1.2)
grid on
title('Carrier Signal f_0 = 5 Hz (Logic 0)')
xlabel('Time (s)')
ylabel('Amplitude')
% 4. FSK Signal
subplot(4,1,4)
plot(total_time,fsk,'LineWidth',1.5)
grid on
title('FSK Modulated Signal')
xlabel('Time (s)')
ylabel('Amplitude')

```

Matlab Output:

Figure 1 x



Discussion/Conclusion:

In this experiment, I successfully implemented FSK modulation and the FSK waveform was observed on DSO and spectrum analyzer. It was verified that two different carrier frequencies are used to represent Logic 0 and Logic 1. The FSK waveform was also generated using MATLAB and the results were verified.

Shu 18/8/26
Signature of faculty in-charge

Dhanya Shah

160103240164

SCALE :-

x-axis: Amp

y-axis: Unit Period

2 Unit = 5V

1 Unit = 14ns

Modulating
Amp →

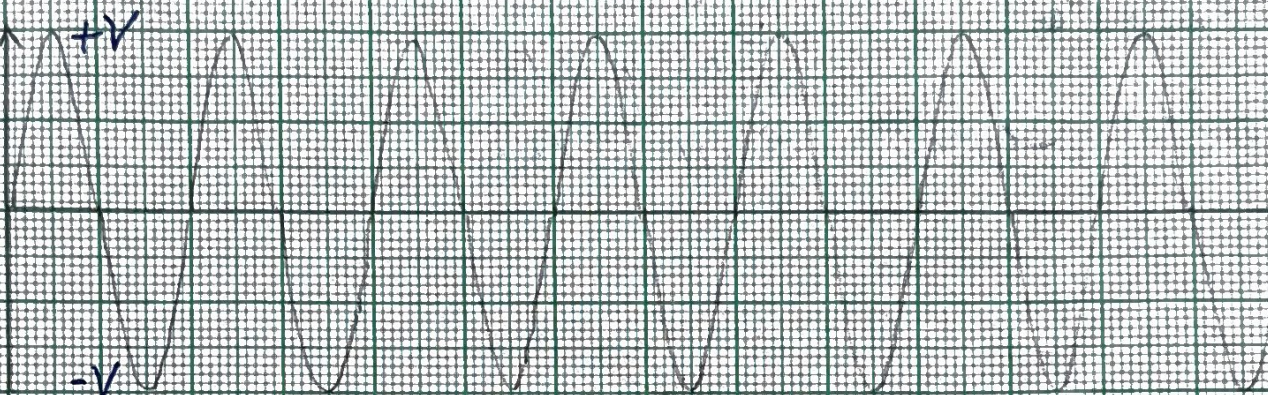
1 0 1 0 1 0 1
+V = 5V

← T_b →

Sq Wave: 1 kHz

T_b

Carrier ①
Amp →



T_b

Series: 4.5 / 5 kHz

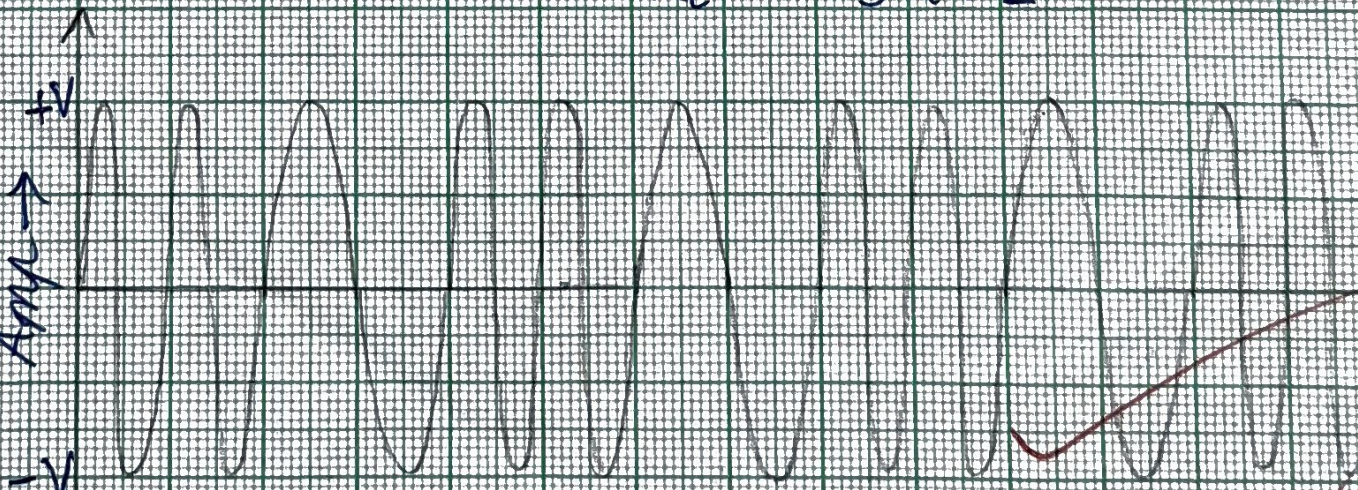
Carrier ②
Amp →



T_b

Series: 13 kHz

Modulated
Amp →



T_b

FSK

Shah
4/18/26